PSI Center for Scientific Computing, Theory and Data

LESTO TASK 3.4 | REPORT

The LESTO case ladder, explained

From the T-Flows benchmark to conservative wall exchange: model equations, verification and experimental comparison

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Cases 001–011: Bojan Ničeno • Cases 012–024: Jan Kren

Purpose of this report This report summarises my recent work on LESTO, building on your carrier and transport cases 001–011. Cases 012–024 extend that foundation with wall exchange, equilibrium chemistry and comparisons with the experiments and T-Flows results in the Lobresco paper [1]. The central questions are which equations have changed, how conservation is enforced, and what the comparisons actually establish. Case 022 checks reproduction of the T-Flows temperature model; case 023 tests the extended physical model against five experiments. These are different tests, not a ranking of CFD codes. The reported onset RMS error is 18 K versus about 35 K for the published models, conditional on the reconstructed experimental inputs.

1 The question

Lead–bismuth eutectic (LBE) is the liquid-metal coolant of some fast reactor designs. Fission products dissolved in it, iodine among them, can evaporate, and iodine does not leave as pure iodine: it combines with lead and bismuth into gases such as PbI_2 and BiI_3. Which compound forms matters, because each one condenses at a different temperature, and therefore ends up in a different place.

To study this, radiochemists use **thermosublimatography**: a long, narrow tube (a few millimetres wide, about a metre long) whose wall temperature falls steadily from about 1000 K at one end to near room temperature at the other. A small boat of material is heated near the hot end; a slow stream of helium carries the vapour downstream; each compound sticks to the wall where the wall is cold enough for it. Afterwards the tube is scanned, and the position of each deposit tells you its **deposition temperature**, T_{dep}.

The coupled problem is: the gas *flows* along the tube, the vapour *diffuses* sideways to the wall, and at the wall it *condenses* or *re-evaporates* depending on the local wall temperature and on what is already deposited there. The goal of the LESTO work in Task 3.4 is a computational fluid dynamics (CFD) model, built on OpenFOAM, that does all three and can be checked against the experiments.

Reading guide. The next section defines the equations and sign conventions. The ladder then identifies the case that checks each added capability. The final sections separate numerical verification, reproduction of the T-Flows benchmark and comparison with experiments. Here “frozen carrier” means that the helium fields are computed first and held fixed while the dilute reactive species evolve.

2 Equations behind the added physics

Transport on the frozen helium field

The steady carrier provides ρ, $\mathbf{u}$, T and p. Each trace gas is advanced with

$$\underbrace{\frac{\partial(\rho Y_i)}{\partial t}}_{\text{storage}} + \underbrace{\nabla\cdot(\rho\mathbf{u}Y_i)}_{\text{advection}} = \underbrace{\nabla\cdot(\rho D_i\nabla Y_i)}_{\text{diffusion}} + S_i^{\text{rel}} + S_i^{\text{wall}}. \tag{1}$$

Y_i is species mass per unit helium-carrier mass; D_i is the molecular diffusivity. The two sources represent release from the boat and exchange with the wall, in $\mathrm{kg\,m^{-3}\,s^{-1}}$. The reactive Y_i do not have to sum to one. Their evolution does not feed back into carrier momentum or temperature. This is the trace-species approximation, and it is least secure close to a strongly evaporating boat.

From a prescribed onset to reversible exchange

The temperature model activates deposition using an imposed threshold. The HKS model instead compares the local vapour pressure with the equilibrium pressure of the condensed phase:

$$j_i = G_i[p_{\text{eq},i}(T_w) - p_{i,w}], \qquad G_i = C_e s_i \frac{2\alpha_i}{2-\alpha_i}\sqrt{\frac{M_i}{2\pi R T_w}}. \tag{2}$$

Here j_i is mass flux in $\mathrm{kg\,m^{-2}\,s^{-1}}$, positive *from the deposit into the gas*; G_i is in $\mathrm{s/m}$. M_i is molar mass in $\mathrm{kg/mol}$, R the gas constant, T_w interface temperature in K, and both pressures are in Pa. α_i is accommodation; C_e and s_i are dimensionless multipliers. The physical baseline sets all three to one.

If $p_{i,w} > p_{\text{eq},i}$, then $j_i < 0$ and vapour condenses. If $p_{i,w} < p_{\text{eq},i}$, then $j_i > 0$ and existing material evaporates. A bare wall cannot evaporate: release is capped by the available inventory. The equilibrium-pressure table selects the stable solid or liquid branch. Thus deposition temperature is an *output*, rather than an imposed threshold. The pressure law alone does not determine the profile: the gas must also reach the surface by diffusion.

Connecting the cell centre to the wall

The finite-volume unknown is $Y_{i,c}$ at the cell centre, not the wall pressure in Eq. (2). Define $\chi_{i,c} = \rho_c R T_c / M_i$, so that $p_{i,c} = \chi_{i,c} Y_{i,c}$. With wall distance d_f and face diffusivity $(\rho D_i)_f$, eliminating the unresolved wall concentration gives

$$\frac{1}{G_{i,f}^{\text{eff}}} = \frac{1}{G_i} + \frac{\chi_{i,c} d_f}{(\rho D_i)_f}, \qquad j_{i,f} = G_{i,f}^{\text{eff}}[p_{\text{eq},i}(T_w) - \chi_{i,c} Y_{i,c}]. \tag{3}$$

The two terms are kinetic and half-cell diffusion resistances in series. Even infinitely fast surface kinetics leave a finite diffusion limit. This is a connection to the resolved gas field, not an extra fitted sticking parameter. Case 016 checks precisely this change.

In that synthetic case the diffusion conductance is $D/d_f = 1\,\mathrm{m/s}$, against a kinetic velocity $G_i\chi_{i,c}/\rho_c = 63.7$–$107\,\mathrm{m/s}$. The combined value is only 0.985–0.991 m/s. Yet the 1 % onset moves by just 0.005 K, while the peak drops by 7.4 % and moves one bin downstream. An unchanged onset therefore does not imply an unchanged deposition profile.

Meaning of the plug-flow reference

Here “plug flow” denotes an idealised, cross-sectionally uniform gas stream, not a two-phase flow regime. The reference calculations follow cooling and saturation without resolving the radial velocity or concentration fields. For case 014, the onset reference is the temperature where the imposed vapour partial pressure meets the equilibrium vapour pressure. The multispecies reference additionally follows equilibrium with fractional removal of condensate. These are simplified numerical checks, not a second CFD solution.

The CFD model transports dilute vapour in helium and stores condensed material in stationary wall inventories. It includes phase change, but does not solve two moving phases: there are no transported droplets, aerosol particles or flowing liquid films.

Conservation during wall exchange

For a single gas–condensate pair, the same realised face flux updates both inventories:

$$S_{i,c}^{\text{wall}} = \frac{1}{V_c} \sum_{f \in c} A_f j_{i,f}, \qquad \Delta m_{i,f}^{\text{wall}} = -\Delta t \, A_f j_{i,f}. \tag{4}$$

A_f is wall-face area and V_c cell volume. The flux used here is the one actually realised by the gas solve after inventory bounds have been applied. Independently updating the wall with a predicted flux would not guarantee cancellation.

Once reactions change molecular identities, conservation must be checked for each element $k \in \{\text{Pb}, \text{Bi}, \text{I}\}$:

$$N_k^0 + N_k^{\text{rel}} + N_k^{\text{in}} - N_k^{\text{gas}} - N_k^{\text{wall}} - N_k^{\text{out}} = 0. \tag{5}$$

All N_k are amounts of the element in mol; inlet and outlet terms include the accounted boundary transport. The ledger reports the residual relative to the case's reference inventory. Round-off closure verifies bookkeeping of the represented processes. It does not establish the accuracy of transport, thermodynamic data or experimental assumptions.

Gas equilibrium preserves atoms, not molecules

For gas species i with ν_{ki} atoms of element k, local element concentrations are

$$b_k = \sum_i \nu_{ki} \frac{\rho Y_i}{M_i}. \tag{6}$$

The equilibrium step redistributes those atoms between the nine gases Pb, PbI, PbI_2, Bi, Bi_2, BiI, BiI_3, I and I_2:

$$\frac{p_i}{p^\circ} = K_{f,i}(T) \exp\left(\sum_k \nu_{ki} \lambda_k \right), \qquad \sum_i \nu_{ki} \frac{p_i}{RT} = b_k, \quad p^\circ = 10^5 \,\text{Pa}. \tag{7}$$

$K_{f,i}$ comes from the thermodynamic tables and the dimensionless element potentials λ_k are solved to satisfy the atom balances. The conversion back is $Y_i = M_i p_i / (\rho RT)$. This assumes instantaneous gas equilibrium, not measured reaction kinetics. In the implemented model the redistribution is requested at $T \geq 350\,\text{K}$; colder cells retain the transported composition. Homogeneous nucleation and aerosols are excluded.

Case 021 also adds shared metal deposits and the forward wall channel

$$3\text{BiI}(g) \longrightarrow 2\text{Bi}(\text{condensed}) + \text{BiI}_3(g). \tag{8}$$

Both sides contain three Bi and three I atoms. Different channels draw on the same physical Bi inventory. The forward-only channel is a modelling choice; it should not be read as a complete reversible surface mechanism.

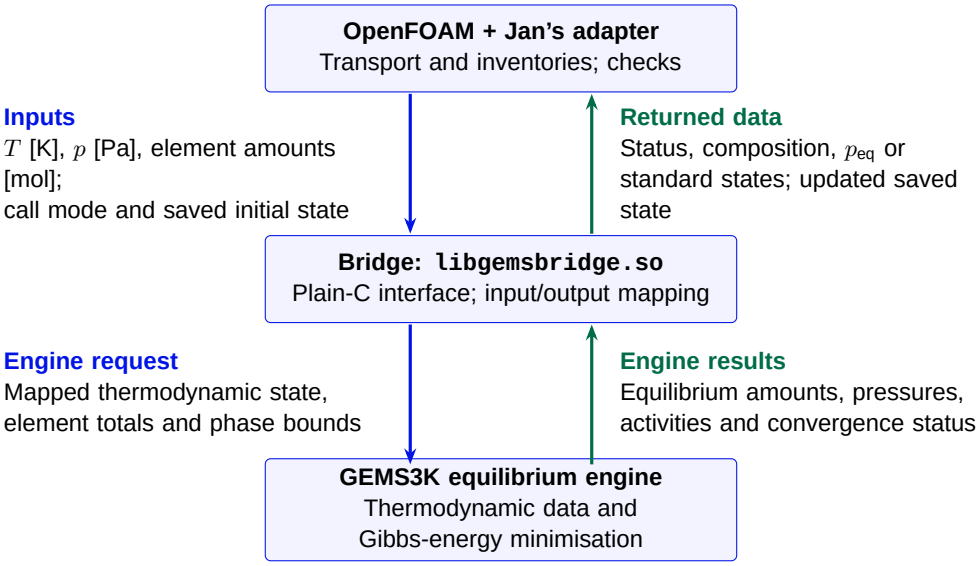
OpenFOAM–GEMS3K coupling

Jan Kren

Jan implemented the optional GEMS3K coupling in the OpenFOAM solver through a compiled bridge library. OpenFOAM advances transport and manages the gas and wall inventories; GEMS3K supplies thermodynamic information. The adapter maps species and elements, passes temperature, pressure and composition, and checks returned states before they are used.

There are two distinct uses. In case 015, local GEMS3K calculations supply $p_{\mathrm{eq},i}$ for the HKS wall and boat exchange. OpenFOAM then evaluates the exchange flux and updates both inventories conservatively. Temperature, dilution, convergence and table-deviation guards select the tabulated pressure when a GEMS result is unsuitable.

In case 020, GEMS3K supplies gas standard states. The *frozen* mode caches formation constants at the prescribed carrier conditions and uses the element-conserving equilibrium kernel each step. The *local* mode also calls GEMS3K per cell and step with a saved initial state; the exact kernel refines the result and provides fallback, enforcing the transported element amounts. Chemistry is therefore coupled sequentially to transport, with no feedback to the frozen helium flow. Cases 015 and 020 verify these optional paths; the main benchmark chain does not require a live GEMS3K call. This coupling is Jan's implementation contribution, distinct from the existing OpenFOAM and GEMS3K software.



Communication during a local equilibrium call. Arrows represent function calls and returned data in memory, not files written at each step. The adapter requests the quantities needed by the selected mode: case 015 uses equilibrium pressure for wall exchange; case 020 uses gas thermodynamics and equilibrium information. In frozen mode, standard states are queried and cached at initialisation, without a local GEMS3K minimisation at every step. GEMS3K does not advance the flow or directly change the wall inventory.

What the bridge consists of

The bridge is a compiled Linux shared library, `libgemsbridge.so` (a `.so` serves the role of a Windows `.dll`). OpenFOAM links to it and exchanges numbers through function calls in memory. It does not write a transfer file at every chemistry step.

File	Role
`gemsbridge.h`	Plain-C interface header: declares the callable functions, input units and return codes.
`gemsbridge.cpp`	C++ source implementing the adapter to GEMS3K.
`libgemsbridge.so`	Compiled shared library loaded with the OpenFOAM solver; built from the bridge source and linked with GEMS3K.
`gemsEquilibrium.C`	OpenFOAM-side adapter: maps fields and species, makes bridge calls and applies checks and fallbacks.
`PbI2He-dat.lst`	Example thermodynamic input manifest: lists the files used to initialise the PbI_2–He chemical system.
`*-dch.dat`, `*-ipm.dat`, `*-dbr-0-0000.dat`	Chemical-system data, equilibrium-solver settings and initial node state, respectively. JSON equivalents are also provided.

The bridge source and header are in `thermochemistry/gemsbridge/`; the example input system is in `thermochemistry/systems/PbI2He/`. The shared library is generated by the build and is not a tracked data file.

One chemistry call OpenFOAM supplies T [K], p [Pa] and element amounts [mol] to `gemsb_equilibrate`. The bridge runs GEMS3K and returns a status; additional calls retrieve species amounts and thermodynamic results. The OpenFOAM adapter checks these results before updating chemistry or using the equilibrium pressure in the wall law. The `.lst` and associated data files initialise the engine; the evolving state is passed in memory.

The plain-C boundary avoids sharing C++ objects between the different compiler/runtime versions used for OpenFOAM and GEMS3K. Each MPI process has its own engine; the bridge itself performs no MPI communication.

3 The case ladder

A model like this has many ingredients: the gas flow, heat, variable gas properties, several chemical species, wall chemistry, gas-phase chemistry, and the real geometry and temperatures of each experiment. If you add them all at once and the answer is wrong, you do not know which ingredient is to blame. So we did the opposite. The folder `run/` of the LESTO repository holds 24 *cases*, numbered 001 to 024, and every case obeys the same four rules:

1. Its `readme.md` starts with *"This case is based on case N"*.
2. It changes **one thing**, and shows the changed lines.
3. It lists the **numbers to check** (e.g. "mass closure below 10^{-15}").
4. It says how to run it; cases 012–024 run with `./Allrun` and print a single word: `PASS`, `FAIL` or `NOTRUN`.

Figure 1 shows the result. The ladder climbs in four stages, which the next sections walk through: the helium flow (001–007), moving vapour through it (008–011), letting the vapour interact with the wall (012–021), and finally the comparisons with experiments and with another code (022–024). Anyone can enter at any rung, rerun it, and see the same numbers.

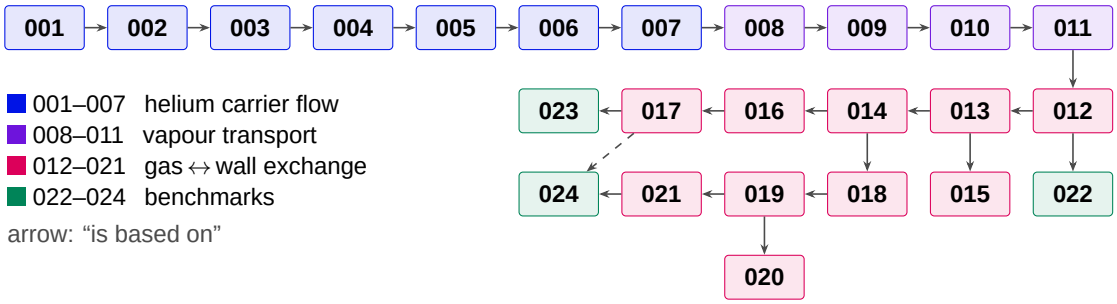


Figure 1: The 24 rungs. The trunk runs left to right along the top, then turns back along the second row. Side branches (015, 018–021, 022) add options that the main line does not need; 024 takes its chemistry from 021 and its realistic helium flow from 017 (dashed).

4 Climbing the ladder

Rungs 001–007: get the helium flowing

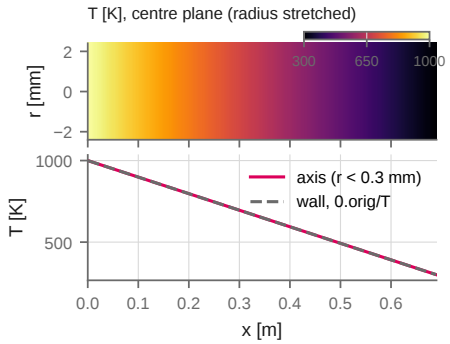
Bojan Ničeno

The first case is the plainest possible flow: helium through a pipe of 2.4 mm radius and 0.69 m length, at room temperature, on a mesh of 228,960 cells. Each next case adds one realistic touch: a parabolic velocity profile at the inlet (002) and inside the pipe from the start (003, which cuts the solver iterations from 388 to 222), the energy equation (004), a wall temperature that falls along the tube (005), a density that follows the ideal-gas law (006), and finally temperature-dependent viscosity and conductivity with a wall falling from 1000 K to 300 K (007, Fig. 2a). The saved result of 007 is the helium "carrier" that the next rungs read.

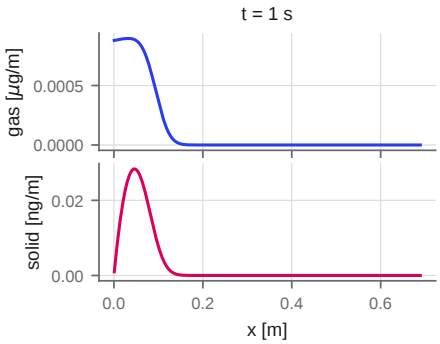
Rungs 008–011: move vapour through it

Bojan Ničeno

Now the flow is frozen and a new solver, `rhoFixedFlowFoam`, moves trace species through it. Case 008 sends PbI_2 vapour and an inert tracer with identical settings and checks that they stay identical (they do, to the last digit). 009 gives PbI_2 its real, temperature-dependent diffusivity, up to 25 times the tracer's on the hot side. 010 introduces a *solid* species, the deposit, and the interface through which chemistry will later talk to the flow. 011 couples them: what leaves the gas appears as solid on the wall, and the two sources add up to exactly zero (Fig. 2b).



(a) 007: the helium carrier. Top: temperature in the tube (radius stretched); bottom: the gas on the axis follows the wall from 1000 to 300 K.



(b) 011: after 1 s, PbI_2 gas enters from the left (top) and turns into deposit on the wall (bottom), mass for mass.

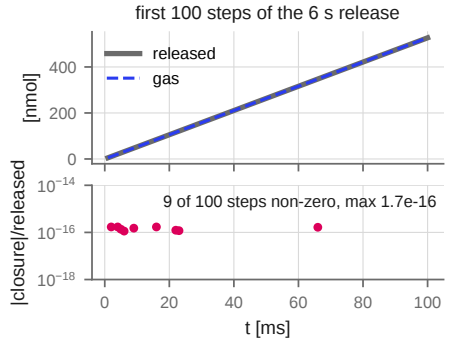
Figure 2: The first two stages: a correct carrier, then vapour that moves and deposits. (a) is drawn from the saved carrier of 007; (b) is a rerun of 011 to 1 s, made on 15 October 2026.

Rungs 012–021: wall exchange and chemistry

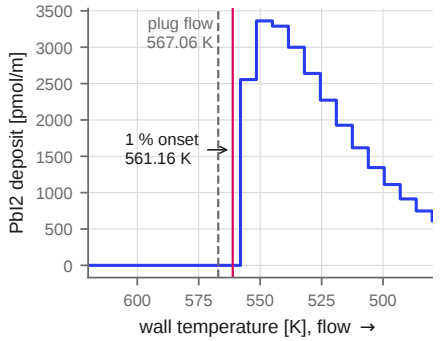
Jan Kren

Up to 011 the deposition rule was a placeholder. From 012 on, the cases add real wall physics, again one piece at a time:

- **012** The simple *temperature model* of Lobresco et al. [1]: vapour sticks once the wall is colder than a set T_{dep}. It also introduces a vapour source (a "boat") and the **mass ledger**: every step, every kilogram is accounted for. The unexplained remainder stays at about 10^{-16} of the released amount, i.e. at the limit of computer arithmetic (Fig. 3a).
- **013** The physical wall law (HKS): condensation and re-evaporation driven by the equilibrium vapour pressure of solid and liquid PbI_2, and a boat that evaporates its real inventory.
- **014** A cheap synthetic test channel (1000 K to 350 K) that runs in 10 s. Here the model's T_{dep} can be checked against the idealised saturation reference: 561.2 K against 567.1 K, within the 10 K gate (Fig. 3b).
- **016** Equation (3) adds diffusion across the unresolved cell-centre-to-wall distance in series with surface kinetics. This resistance controls uptake in this test; the half-cell treatment is also used in the physical benchmarks 017, 023 and 024.
- **017** The real helium flow and measured wall temperature of one Lobresco experiment, instead of the synthetic channel.
- **018–021** More chemistry: a second iodide (BiI_3), nine Pb–Bi–I gases that re-balance with each other everywhere in the tube (019; 020 does the same with GEMS3K), and metal vapour from LBE with shared deposits and wall reactions (021). In 021, all five deposit onsets land within 3.9 K of the simplified reference calculation.



(a) 012: released and gas inventory (top) match; the unexplained remainder (bottom) is $\sim 10^{-16}$, machine round-off.



(b) 014: PbI_2 deposit along the synthetic channel. The model's onset (1 % rule, 561.2 K) sits 5.9 K from plug flow.

Figure 3: Two kinds of check used throughout the ladder: nothing is lost (a), and the answer agrees with a simplified reference calculation where one exists (b).

5 The top rungs: does it match reality?

The last three cases do not add physics; they *test* it.

022 — reproduction of the T-Flows benchmark. This case returns to the temperature-model calculations in Lobresco's master's work with Bojan, published in Ref. [1], using Bojan's T-Flows code. In "paper mode" our model repeats their calculation: the gas level agrees with their Fig. 9 to 0.5 %, and all ten lines of their Table 2 agree within the resolution of the wall temperature table. The one visible difference, a deposit starting 2.6 cm earlier, disappears (to 0.9 mm) with the T-Flows parameter set; confirming which set generated the published figures is a specific point to resolve with Bojan using the original T-Flows inputs. The remaining 0.9 mm is an onset-position comparison, not proof that the entire profiles coincide.

023 — against five experiments. The five PbI_2 experiments of Lobresco et al. [1] differ in helium flow (5, 106 and 540 mL/min) and duration (6 to 300 s). We ran all five with the HKS wall law, the measured wall temperatures, and *no parameter adjusted to the data*, then applied the paper's own rule for T_{dep} (the point where the deposit reaches 1 % of its peak). Table 1 and Fig. 4a show the outcome: an RMS error of 18 K, against about 35 K for each of the two published model configurations. This is a comparison of complete modelling setups, not an isolated comparison of the OpenFOAM and T-Flows discretisations. A coarser mesh changes the answers by at most 0.3 K, and the mass ledger closes to 3.3×10^{-16}.

For a deposit profile $q(x)$, the reported onset is the first crossing of $0.01 \max q$ in the selected deposit region:

$$T_{\mathrm{dep},r} = T_w(x_{1\%,r}), \qquad E_{\mathrm{RMS}} = \sqrt{\frac{1}{5}\sum_{r=1}^{5}(T^{\mathrm{model}}_{\mathrm{dep},r} - T^{\mathrm{exp}}_{\mathrm{dep},r})^2}. \tag{9}$$

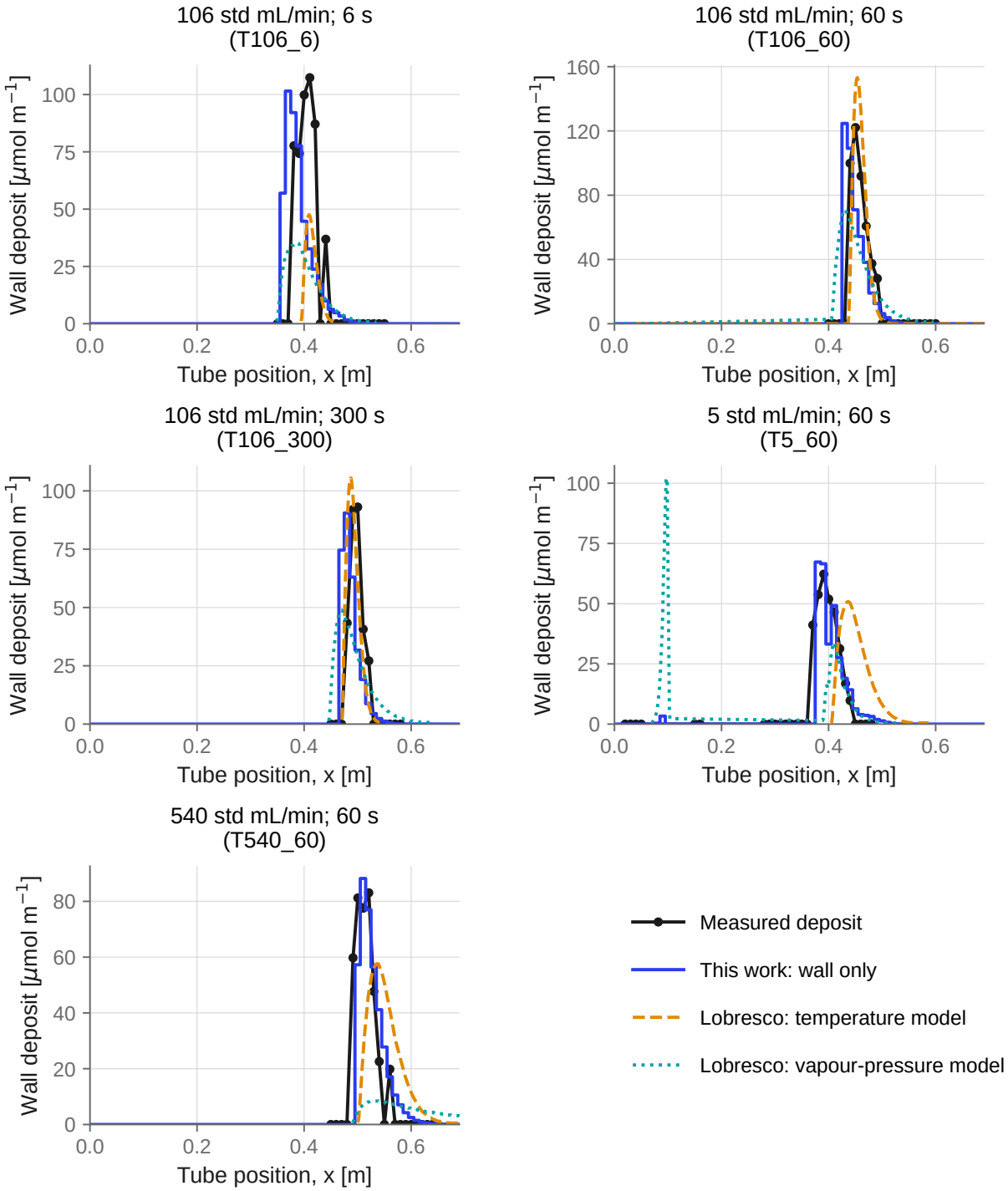
It is an onset statistic, not a peak-position or full-profile error. The low-flow case uses the main downstream deposit, as noted below.

Table 1: Case 023: predicted minus measured deposition temperature, production mesh. Experiment names: helium flow (mL/min) _ duration (s).

	T106_6	T106_60	T106_300	T5_60[a]	T540_60	**RMS**
This work (023)	$+26$ K	$+19$ K	$+16$ K	-11 K	-14 K	**18 K**
Lobresco et al., both models						35 K

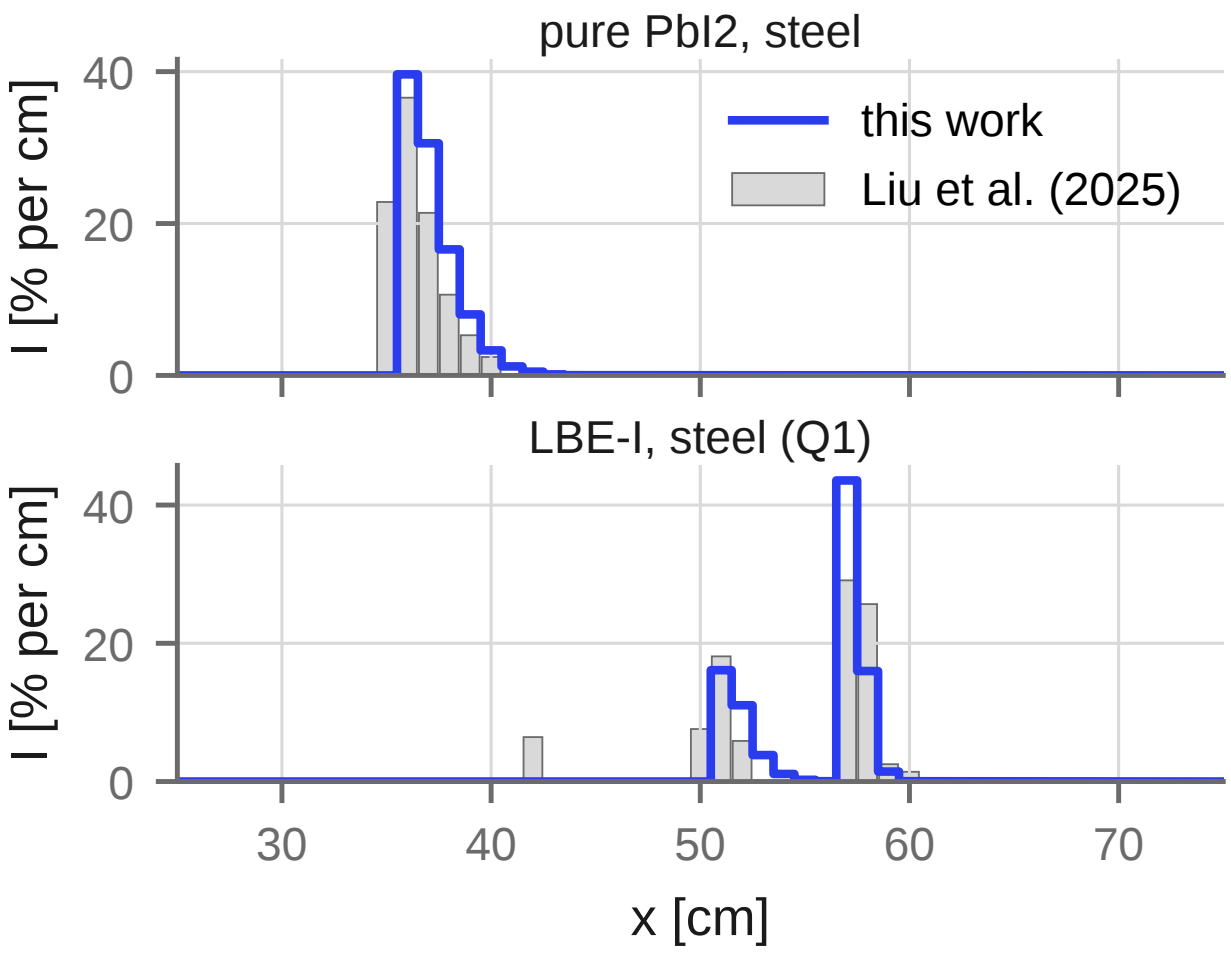
[a] At 5 mL/min some vapour diffuses upstream and leaves a small deposit near the cold inlet; the value is for the main deposit downstream of the boat.

024 — the full Pb–Bi–I chemistry. Liu et al. [2] evaporated pure iodides and iodine-containing LBE into helium in steel and silica columns. Case 024 runs their configurations with the full chemistry of 021 on the realistic flows of 017, 60 s each on a mesh of 41,472 cells. For pure PbI_2 on steel the computed peak sits at 36 cm, 366.5 °C, against the measured 368 ± 18 °C, and for iodine from LBE the model reproduces the two separate deposit bands (Fig. 4b). Refining the mesh to 93,312 cells moves the profiles by 1.98 %.



(a) Case 023: wall-deposit profiles for the five experiments of Lobresco et al. [1]. Blue curves use only condensed wall material at the end of each run; residual gas is excluded. The 5-degree wedge inventory is multiplied by 72 to represent the full column. Each plot has its own vertical range.

Figure 4: The benchmarks, panel (a). Black dots: measurements from Lobresco et al. [1]. Blue: this work. Dashed orange and dotted teal: the temperature-based and vapour-pressure-based models, respectively, of Lobresco et al. [1]. Titles give helium flow at standard conditions (273.15 K, 101325 Pa) and run duration: T106_6 means 106 standard mL/min for 6 s. These five experiments all belong to ladder case 023; panel (b), on the next page, shows case 024.



(b) Case 024: two steel-column configurations from Liu et al. [2], with pure PbI_2 as source (top) and iodine-containing LBE as source (bottom). The ordinate is the iodine yield in each 1 cm bin. Both are baseline Q1 configurations.

Figure 4: The benchmarks, panel (b), continued. Grey bars: measurements from Liu et al. [2]. Blue: this work.

6 What this gives us, and what it does not

What we have

- **A model with traceable verification checks.** Every capability of the solver has a rung that isolates it and a number that checks it. When something breaks later, `./Allrun-ladder 012-024` reruns all checks and points at the first rung that fails.
- **Nothing goes missing.** The recorded pair and element ledgers close near round-off in the cited cases. These checks address conservation, not the adequacy of the physical model.
- **A useful prediction.** Against five experiments the untuned model halves the error in T_{dep} of the published models (18 K vs 35 K RMS), and it reproduces the multi-band deposits of the LBE experiments.
- **A paper trail.** Cases 023 and 024 feed directly into the methods manuscript (Sec. 6.3, Table V, Fig. 5) and the ladder is the availability section of LESTO deliverable D38.

What we do not have yet

- Some inputs of the experiments are not published and are *assumed*: how fast the boat released its material, the reference conditions of the flow, and several positions. The 024 results are therefore labelled provisional.
- The model's deposit peaks sit 1 to 4 cm upstream of the measured ones in 023; the boat's obstruction of the flow and the dense vapour right next to it are not represented.
- The BiI_3 cases (018, 019, 021) need restricted thermodynamic tables that cannot be redistributed, and 015 and 020 need an optional GEMS3K build; without them those cases print `NOTRUN`.
- The mesh study of 024 shows small changes (1.98 %) but no formal grid-convergence index: the apparent orders of convergence are not consistent.

How to try it Every case lives in `run/<NNN-name>/`. Read its `readme.md`, then run `./Allrun` (cases 012–024; outputs go into a git-ignored `work/` folder) and look for PASS. The cheap cases take seconds to minutes; 017 about 8 minutes; the full 023 and 024 studies take hours per experiment. The slide-by-slide version of this report is `presentations/case-ladder/lesto-case-ladder.pdf`.

References

[1] F. Lobresco, G. Tiebel, G. C. Conigliaro, P. Steinegger, O. Marinich, N. I. Prasianakis, J. Neuhausen, B. Ničeno, *Assessment of temperature-based and vapor-pressure-driven models for CFD simulation of volatile species deposition*, Nucl. Eng. Des. 459 (2026) 115201. doi:10.1016/j.nucengdes.2026.115201

[2] L. Liu, V. Zobnin, I. Zivadinovic, R. Eichler, A. Vögele, D. Herrmann, A. Aerts, P. Steinegger, J. Neuhausen, *Chemical speciation of iodine evaporated from liquid lead–bismuth eutectic investigated by thermosublimatography*, J. Radioanal. Nucl. Chem. 334 (2025) 5201–5216. doi:10.1007/s10967-025-10246-4

Nomenclature

Symbols follow Eqs. (1)–(7) and the benchmark definitions. A dash denotes a dimensionless quantity.

Symbol	Meaning	SI unit
A_f	Area of wall face f	m^2
b_k	Gas-phase concentration of element k (amount of atoms per volume)	$\mathrm{mol\,m^{-3}}$
C_e, s_i	HKS prefactor and species kinetic multiplier; baseline values are one	–
D_i	Molecular diffusivity of gas species i in helium	$\mathrm{m^2\,s^{-1}}$
d_f	Distance from cell centre to wall face	m
E_{RMS}	Root-mean-square error of predicted deposition-onset temperatures	K
G_i, $G_{i,f}^{\mathrm{eff}}$	Kinetic and effective wall conductances, respectively	$\mathrm{s\,m^{-1}}$
j_i, $j_{i,f}$	Wall mass flux; positive for evaporation into the gas	$\mathrm{kg\,m^{-2}\,s^{-1}}$
$K_{f,i}(T)$	Formation equilibrium constant of gas species i; here f means formation	–
M_i	Molar mass of species i	$\mathrm{kg\,mol^{-1}}$
$m_{i,f}^{\mathrm{wall}}$	Condensed mass of species i stored on wall face f	kg
N_k	Amount of element k in an inventory or cumulative transfer	mol
p, p_i	Prescribed carrier pressure and species partial pressure	Pa
$p_{\mathrm{eq},i}$	Equilibrium vapour pressure over the condensed phase	Pa
p°	Standard pressure, 10^5 Pa	Pa
$q(x)$	Axial deposit profile used for the relative 1 % onset criterion	Profile-dependent[a]
R	Universal gas constant	$\mathrm{J\,mol^{-1}\,K^{-1}}$
S_i^{rel}, S_i^{wall}	Volumetric species mass sources from the boat and wall	$\mathrm{kg\,m^{-3}\,s^{-1}}$
T, T_w	Gas and wall/interface temperatures	K
T_{dep}	Wall temperature at the selected deposit onset	K
t, Δt	Time and time-step duration	s
u	Helium-carrier velocity	$\mathrm{m\,s^{-1}}$
V_c	Volume of cell c	m^3
x, $x_{1\%,r}$	Axial coordinate and 1 %-of-peak onset position for run r	m
Y_i	Species mass per unit helium-carrier mass	–
α_i	Accommodation coefficient; baseline value is one	–
λ_k	Dimensionless element potential in the equilibrium solve	–
ν_{ki}	Number of atoms of element k in one molecule of species i	–
ρ	Helium-carrier mass density	$\mathrm{kg\,m^{-3}}$
$\chi_{i,c}$	Pressure conversion factor $\rho_c RT_c/M_i$, with $p_{i,c} = \chi_{i,c} Y_{i,c}$	Pa

[a]The criterion uses $q/\max q$, so a uniform change of profile units or normalisation does not change the onset.

Indices and superscripts

i: gas species; k: element (Pb, Bi or I); c: cell centre; f: wall face, except in $K_{f,i}$; w: wall/interface; r: experimental run. Superscripts 0, rel, in, gas, wall and out identify initial inventory, release, cumulative inflow, gas inventory, wall inventory and cumulative outflow. Superscripts model and exp denote prediction and measurement; Δ denotes a change over one step.

Abbreviations and chemical notation

CFD: computational fluid dynamics; **HKS**: Hertz–Knudsen–Schrage; **LBE**: lead–bismuth eutectic; **RMS**: root mean square; **SI**: International System of Units. **GEMS3K** is the chemical-equilibrium kernel used in the optional cases. He, Pb, Bi and I denote helium, lead, bismuth and iodine; subscripts in chemical formulae count atoms. The labels g and condensed denote gas and the immobile condensed reservoir, respectively.